SPE-227109-MS

Post-Stimulation Diffusion Time Assessment on Early Hydrocarbon Breakthrough and Initial Water Recovery of Unconventional Wells

Sohrat Baki, Abdulrahman Alowaid, Fahad Alghamdi, and Rakan Mustafa, Saudi Aramco, Dhahran, Saudi Arabia

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

Abstract

Post-stimulation diffusion time is defined as the time period at which hydraulic fracturing fluids diffuse into the formation after the completion of pumping operations. It is measured from the conclusion of stimulation operations until the well is first opened for post-stimulation flowback operations. The primary objective of this paper is to comprehensively study the impact that diffusion time has on hydrocarbon breakthrough and initial water recovery, and to thoroughly investigate any potential association with overall well performance.

The onset of production from unconventional wells that underwent stimulation through hydraulic fracturing is dominated by high water cut, starting from 100% and gradually dropping with time as the well cleans up and hydrocarbons breakthrough into the production stream. The process followed in this paper starts with a broader review on a large group of wells looking at two main parameters with time: water production rate which is expected to peak at the start of production and progressively drop, and the percentage of water recovery after a fixed period of time. After that, the process takes a closer look into a sample set of closely-placed wells that have similar formation properties, to quantify the impact on well performance, specifically through the use of hydrocarbon production and productivity index normalized by lateral length versus time.

The first part of the paper investigates a wide spectrum of unconventional wells to establish a baseline understanding on the impact of diffusion time. Looking at initial water production rates, both peak water production and water production profiles were lower with higher diffusion time, indicating an inverse relationship between the two. Similarly, Hydrocarbon breakthrough had an inverse relationship with diffusion time, where gas/oil breakthrough was faster when diffusion time was longer. The second part of the paper looked into the impact of diffusion time on well performance, where a group of wells within close proximity were evaluated. Results from that comparison indicated the absence of a correlation between diffusion time and well performance. Results from this study addressed potential concerns related to shutting in wells post-stimulation in the application area, especially on wells within close proximity where formation properties are relatively homogenous.

The novelty of this paper resides in its in-depth examination of a critical transition period between stimulation and production of unconventional wells, which might have been overlooked the past. The approach adopted in this paper can be readily implemented at any unconventional basin, to provide a

comparison with the findings from this paper. This in turn broaden its practical application and relevance to the field of unconventional oil and gas production.

Introduction and Overview

Over the past couple of decades, development activities within unconventional resource (UR) assets have grown exponentially with increasing energy demands. These ultra-low permeability formations require extensive stimulation to maximize the stimulated rock volume (SRV), establishing conductive pathways for commercial hydrocarbon production. Multi-stage hydraulic fracturing operations have become the industry standard for stimulation of UR assets, yielding effective SRV. These operations require large volumes of water and sand to create and propagate effective fractures. Following fracturing, wells are put on production and undergo flowback cleanup operations, during which a significant portion of the previously injected water is recovered before hydrocarbon production begins. The period between the end of fracturing and the start of flowback is referred to as "post-stimulation diffusion time" or "soaking time", typically measured in days. The rationale behind this naming is that it infers the time allowed for pumped fluid to diffuse or soak into formation before production.

This study dives deep into the correlation of post-stimulation diffusion time and well behavior. The assessment is performed on multiple wells within the same unconventional play to ensure a fair and consistent comparison. All wells have been stimulated via multi-stage hydraulic fracturing using a slickwater design, and underwent the same operational sequence as fracturing, coiled tubing plug milling, then flowback testing as illustrated in Figure 1. Diffusion time for all wells was uniformly calculated as the time from the moment the well is shut-in after the final hydraulic fracturing stage, to the time at which the well is open for flowback. The main objective was to analyze if there were any notable observations on well performance, and overall productivity. This was performed by investigating three key parameters: initial water production rate, percentage of water recovery, hydrocarbon breakthrough, and normalized productivity index (PI). The first three parameters will indicate post-stimulation well cleanup efficiency and impact on the onset of hydrocarbon breakthrough, while the last, PI, will assess implications on well performance and deliverability.

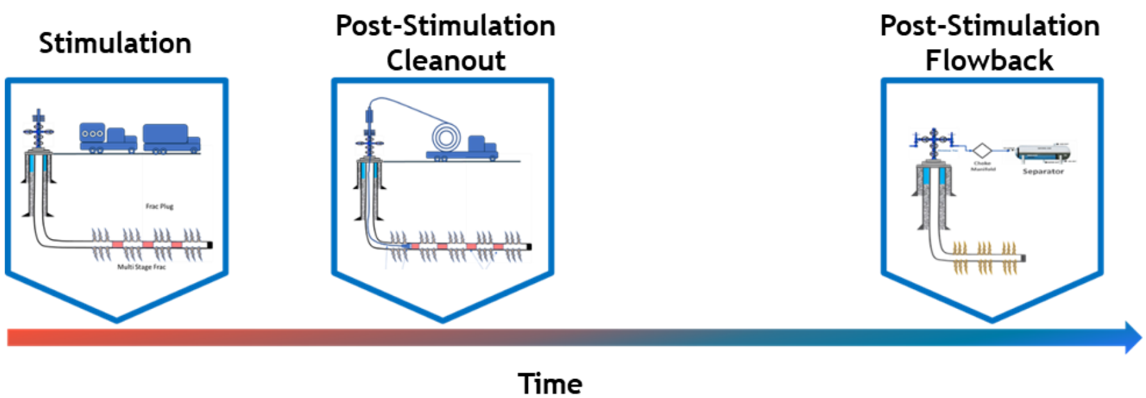

Figure 1—Generic schematic highlighting a typical well lifecycle from stimulation to flowback testing

A review of the literature has revealed several insights related to the subject in various UR plays globally. Yaich et al (2013) highlighted that operators in the Marcellus shale have reported significant production performance improvements after extended shut-in periods, which was attributed to the unavailability of gas pipelines at that time and place. Cheng (2012) indicated that water production reduces with higher diffusion time, ultimately lowering surface water handling costs. This was attributed to the absorption of water in fractures by the rock matrix due to prolonged shut-in periods, increasing relative permeability to hydrocarbons and improving well productivity. Ibrahim et al. (2019) presented case studies investigating the effect of diffusion time on well performance using rate transient analysis (RTA) and decline curve analysis (DCA), before and after production shut-in to evaluate changes in SRV. Their study found that long shut-in periods increase gas flow rates and reduce water production rate.

Methodology

This study utilizes an empirical approach, leveraging data from actual unconventional wells that underwent similar stimulation and flowback operations. A comprehensive analysis of multiple parameters was conducted to extract meaningful correlations, including diffusion time, water cumulative production after a fixed time period, and hydrocarbon breakthrough. All wells considered in this study follow a uniform lifecycle, as illustrated in Figure 1. The lifecycle consists of three stages: (1) Multi-stage hydraulic fracturing; (2) Coil tubing milling operation to remove fracturing plugs used for zonal isolation during stimulation; and (3) well testing and flowback operations for cleanup testing. Results from the latter stage in conjunction with the diffusion time of each well were the foundations of this study.

To investigate the effect of diffusion time, multiple sets of wells were considered, each with varying diffusion times. The differences in diffusion time enables the identification of trends and patterns that can inform post-stimulation well diagnostics and potential optimization opportunities for future well operations. Figure 2 illustrates the distribution of wells into distinct groups, highlighting the number of wells in each group and their corresponding diffusion times, which highlights a right-skewed distribution with most wells located within a shorter diffusion timeframe. Nonetheless, diffusion times ranged from relatively short periods to extended durations, allowing for a comprehensive analysis of its impact on hydrocarbon breakthrough and overall well performance.

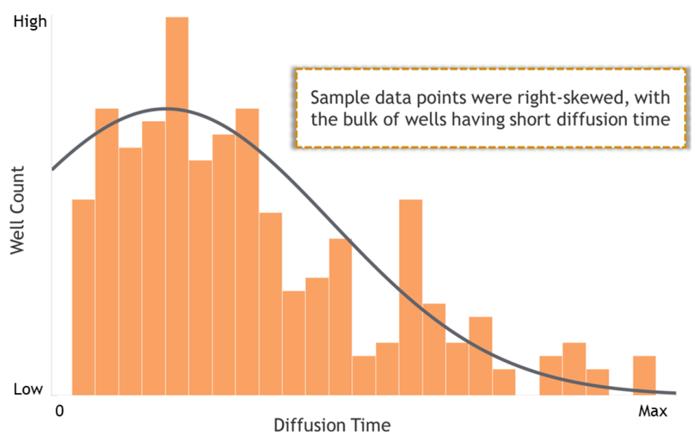

Figure 2—Distribution of well count versus diffusion time

Results and Observations

Impact of Diffusion Time on Initial Water Production

The first section of this study investigates the relationship between post-stimulation diffusion time and initial water production, an essential aspect of unconventional well cleanup. The goal of this assessment is to evaluate the relationship between the two at the onset of production, when wells are dominated by water production and high water cut. The results of this analysis are considered qualitative, focusing on peak water production and water rate profile over time. Diffusion time was categorized into three groups: low, intermediate, and high, with most wells falling into the low category. Figure 3 shows water production rates for several wells over flowback time, grouped by diffusion time category. The results reveal a clear inverse relationship between water production and diffusion time. In other words, shorter diffusion times are associated with higher initial water production, and longer diffusion times with lower initial water production. Wells in blue and green have low diffusion times, those in orange have intermediate diffusion times, and those in red have high diffusion times. The results indicated that longer diffusion times are linked to lower water production rates, and that wells with shorter diffusion times had higher and more rapid water production profiles. Consequently, peak water production followed the same pattern, with wells having low diffusion times exhibiting higher peak water production, and those with higher diffusion times showing lower peak water production.

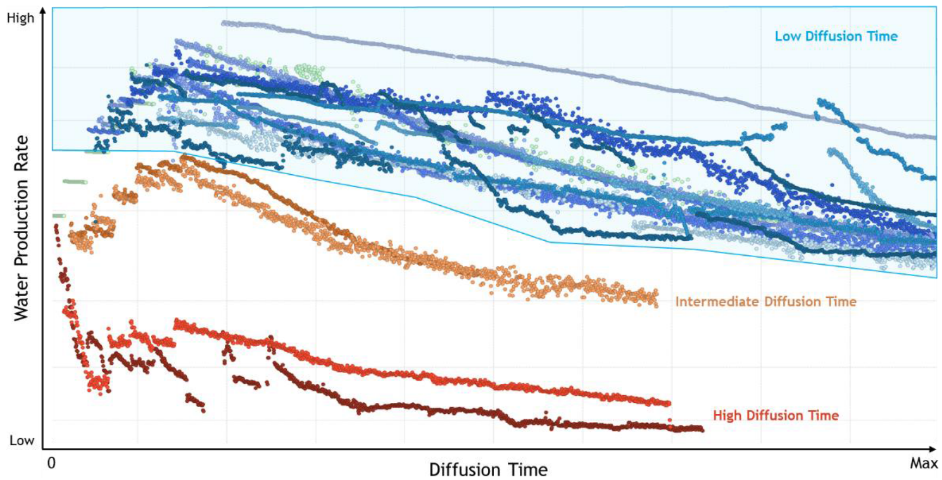

Figure 3—Water production rates vs diffusion time for all three categories

Impact of Diffusion Time on Initial Water Recovery

This section expands on the findings of the previous section, examining the cumulative water recovery after a fixed period, referred to as "X". Figure 4 illustrates the relationship between cumulative water recovery at period "X" and diffusion time, with the data plotted as a scatter plot. Most data points cluster at the lower end of the diffusion time range, consistent with the previous section's finding that most wells have shorter diffusion times. Generally, cumulative water recovery across the entire diffusion time period showed a wide spectrum. However, wells that had shorter diffusion time, exhibited a high variability compared to its long diffusion time counterpart. This observation provides a qualitative assessment into the impact of water diffusion and variability of water recovery. To further support this finding, an exponential decline trendline is fitted to the data, showing that water recovery at period "X" decreases rapidly with increasing diffusion time, consistent with the findings of the previous section of an inverse relationship between the two parameters. This observation suggests that longer diffusion times lead to prolonged cleanup and reduced initial water production.

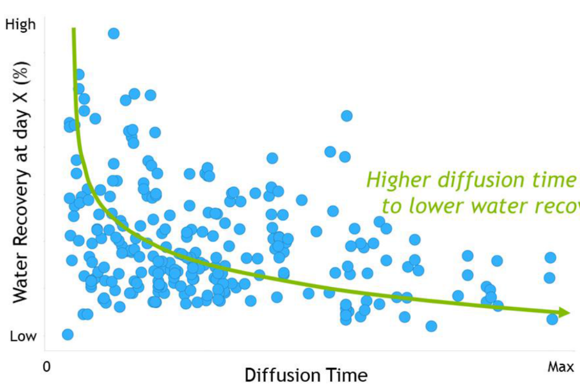

Figure 4—Percentage of water recovery at time period X versus diffusion time

Impact of Diffusion Time on Well Performance (Productivity Index)

Previous sections thoroughly examined the impact of diffusion time on water recovery and well cleanup. These previous findings provided insights into potential improvements in operational efficiency and optimization of activities. On the other hand, a crucial aspect that necessarily requires consideration is the effect of well deliverability, which primarily relates to the overall recovery of subject wells. To evaluate the performance of wells, the study conducted an analysis of PI normalized by effective lateral length for nearby wells that shared similar completion and subsurface properties. As these wells produced all three phases, including gas, condensate, and water, gas-equivalent PI was utilized. Table-1 provides a summary of the results of normalized PI and diffusion time for a fixed production time for four wells located in the same multi-well pad. The wells had different diffusion times, specifically ranging from the shortest to the longest: Well-A, Well-B, Well-C, and Well-D. The results indicate that Well-C and Well-D exhibited higher initial PI at the end of the fixed production time, followed by Well-B, and then Well-A, which suggests a direct relationship between diffusion time and PI for this particular example.

Table 1—Summary of an example-1 of a multi-well pad with variable diffusion times

Well	Diffusion Time	Productivity Ranking
A	Lowest	4th
B	Low	3rd
C	High	1st (equal)
D	Highest	1st (equal)

In contrast, a comprehensive assessment of another multi-well pad yielded conflicting and somewhat unexpected results. The relationship between PI and diffusion time was found to be inconsistent and somewhat ambiguous. As clearly shown in Table 2, wells with relatively low diffusion times had relatively higher PIs, while those with high diffusion times had noticeably lower PIs. The results from these two distinct examples, namely examples 1 and 2, suggest that there is no clear, definitive, or straightforward correlation between PI and diffusion time, which is an important consideration for future studies.

Table 2—Summary of an example-2 of a multi-well pad with variable diffusion times

Well	Diffusion Time	Productivity Ranking
A	Lowest	2nd
B	Low	1st
C	High	4th
D	Highest	3rd

To further generalize these findings, Figure 5 highlights PI at a specific production time "X" against diffusion time. A larger dataset of wells was used to increase the range of investigation. Wells were grouped and color coded, where each color represents an identical set of wells with similar characteristics. The absence of a correlation between diffusion time and PI was again apparent when analyzing a broader range of wells, suggesting that diffusion time is not a correlating factor to well performance in unconventional wells.

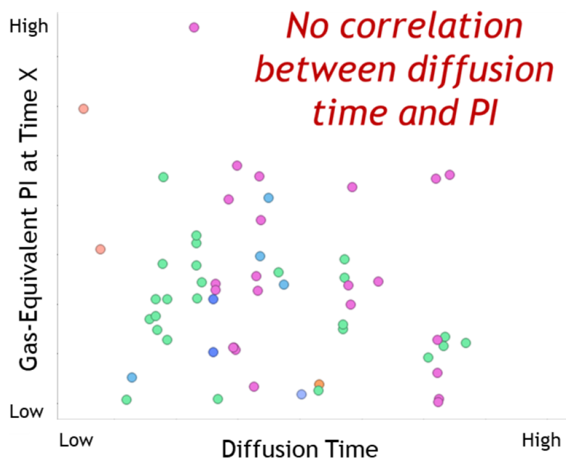

Figure 5—Gas-equivalent productivity index vs diffusion time for color coded cluster of wells

Impact of Diffusion Time on Well Performance (Cumulative Production)

To further investigate the impact of diffusion time on well performance, cumulative production at time "X" was analyzed. Wells were grouped into two categories based on their condensate-to-gas ratio (CGR): low CGR and high CGR. Figure 6 displays a combination chart of cumulative gas-equivalent volume produced at time "X" versus diffusion time for low-CGR wells. The bar chart shows the average cumulative gas-equivalent volume at time X, and the red line displays the well count for each diffusion time range. The results were consistent with previous findings, showing no direct correlation between cumulative production and diffusion time. The mid-range period with higher cumulative production is attributed to faster hydrocarbon breakthrough for wells at that timeframe. The well count distribution is consistent with previous plots, with most wells having shorter diffusion times. The same exercise was completed for the other category with higher-CGR wells, as shown in Figure 7. Results were consistent with the above findings, indicating the absence of a correlation between cumulative hydrocarbon production and diffusion time.

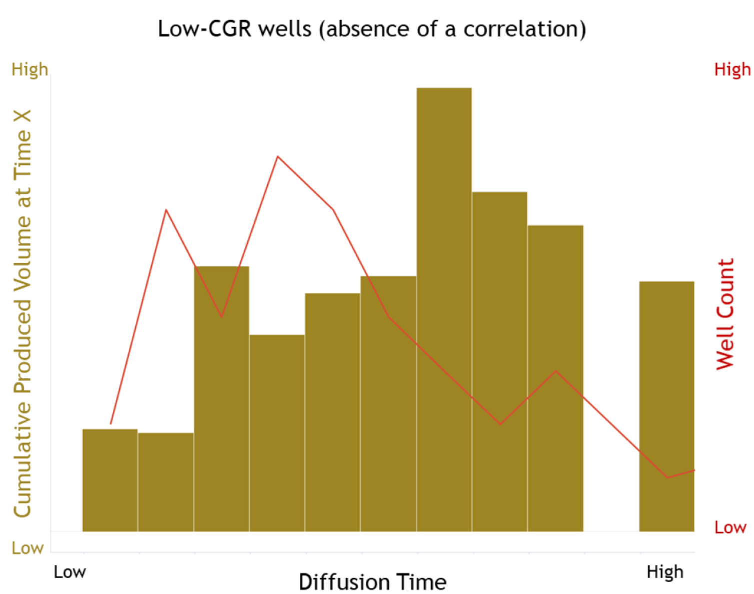

Figure 6—Cumulative produced volume and well count vs diffusion time for low-CGR wells category

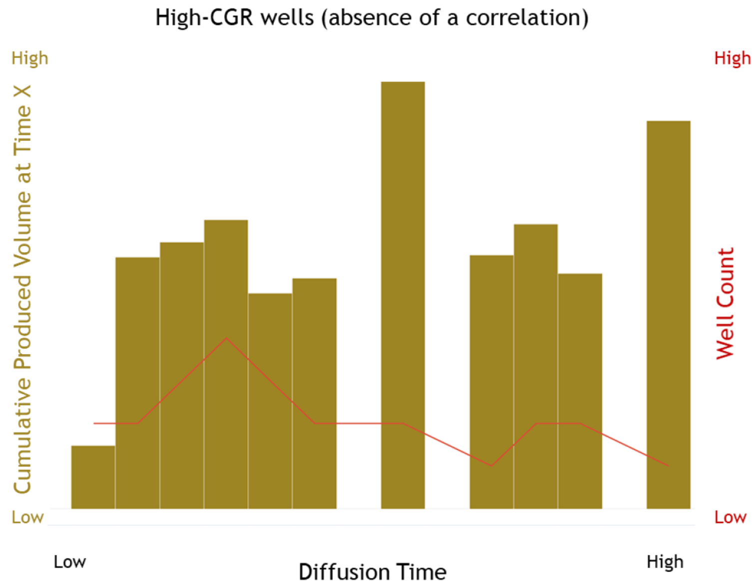

Figure 7—Cumulative produced volume and well count vs diffusion time for high-CGR wells category

In summary, the results of this investigation provided valuable insights into the relationship between post-stimulation diffusion time and well performance in unconventional wells. The analysis reveals that longer diffusion times were associated with lower initial water production rates and faster hydrocarbon breakthrough, supported by a higher initial water recovery. Furthermore, the results indicate that diffusion time has no clear impact on well deliverability, as measured by PI and cumulative produced hydrocarbon volumes, suggesting that other factors such as well placement and completion design could be more influential in influencing well performance. The findings of this study have significant implications on the optimization of unconventional well development and operations, highlighting the need for further research into the relationship between diffusion time and hydrocarbon recovery, considering various factors and conditions, to improve the understanding of these complex systems and ultimately enhance well performance.

Conclusions and Takeaways

The study focused on assessing diffusion time, which is the time from the end of the last hydraulic fracturing until the onset of cleanup flowback. The study looked into the impact of post-stimulation diffusion time on multiple variables, mainly initial water production, overall water recovery, hydrocarbon breakthrough, and well performance. The study was divided into two main parts in relation to diffusion time: the first focused on was through the evaluation of initial water production and water recovery, while the second focused on the impact on diffusion time on well deliverability and productivity. The first part of the study found an inverse relationship between water production rates, cumulative water recovery, and diffusion time. Findings from this section can be summarized in an inverse correlation between diffusion time and water behavior. As diffusion time increases, both initial water production and cumulative water recovery decrease. This is evident in the results, which showed that wells with low diffusion times have significantly higher water production rates than those with high diffusion times. This is attributed to the fact that previously pumped water had less time to leak-off and diffuse into formation. This difference highlights the significant impact of diffusion time on water recovery, suggesting that extending diffusion time could minimize initial water production and accelerate hydrocarbon breakthrough during initial production. By allowing pumped fluids to diffuse into formation before production, cleanup flowback durations can be reduced, further improving overall operational efficiency. Another observation was the earlier hydrocarbon breakthrough, which accelerated three-phase production. This might be desirable in special cases where water handling can be challenging. Diffusion time can be adjusted by analyzing the overall well completion lifecycle scheduling to space out the three main operations: multi-staged hydraulic fracturing, coiled tubing milling operation, and well testing and flowback operations.

The second part of the study focused on assessing potential impacts on well performance by evaluating the relationship between PI and diffusion time. The findings from this study indicated the absence of a direct correlation between the two parameters, which concluded that there wasn't a direct negative impact on the performance of studied wells. Similarly, there was no direct correlation between the cumulative hydrocarbon production at a specific time period and diffusion time, even when looking at fields with varying CGRs. This indicates that other factors have greater impact on well productivity, which provides opportunities for future research. To further understand potential implication of diffusion time on well performance, additional parameters should be introduced into the study such as the area, field, formation, fluid type, petrophysical properties, and stimulation strategy. The findings of this paper were limited to the set of wells analyzed, and should be expanded in the future with additional data to confirm these outcomes and identify any abnormalities.

References

Cheng, Y. (2012, February 2012). "Impact of Water Dynamics in Fractures on the Performance of Hydraulically Fractured Wells in Gas-Shale Reservoirs". *Journal Paper*: 143–151. Paper Number: SPE-127863-PA. https://doi.org/10.2118/127863-PA

Rong, Lu "Investigation of Capillary Pressure Effects on Well Cleanup after Hydraulic Fracturing", Rong Lu, Colorado School of Mines. 2012 Arxiv

Masoud Alfi, Bicheng Yan, Yang Cao, Cheng An, Yuhe Wang, John Killough. Three-Phase Flow Simulation in Ultra-Low Permeability Organic Shale via a Multiple Permeability Approach" Paper presented at the SPE/AAPG/SEG Unconventional Resources Technology Conference, Denver, Colorado, USA, August 2014. Paper Number: URTEC-1895733-MS. https://doi.org/10.15530/URTEC-2014-1895733

E. Yaich, S. Williams, A. Bowser, P. Goddard, O.C. Diaz de Souza, R.A. Foster "A Case Study: The Impact of Soaking on Well Performance in the Marcellus". Paper presented at the SPE/AAPG/SEG Unconventional Resources Technology Conference, July 20-22, 2015. URTEC-2154766-MS. https://doi.org/10.15530/URTEC-2015-2154766

Ahmed Farid Ibrahim; Mazhar Ibrahim; Pieprzica Chester "Developing New Soaking Correlation for Shale Gas Wells". Paper presented at the SPE/AAPG/SEG Unconventional Resources Technology Conference, Denver, Colorado, USA, July 2019. Paper Number: URTEC-2019-393-MS. https://doi.org/10.15530/urtec-2019-393